

Secure Cloud-Native Application Deployment on AWS using Kubernetes (EKS)

Course: CS581 — Cloud Security Engineering **Project:** Signature Project **Application:** MuscleQuant AI — surface EMG monitoring for athletes **Authors:** Emmanuel Owusu, Gelin Deng

1. Executive summary

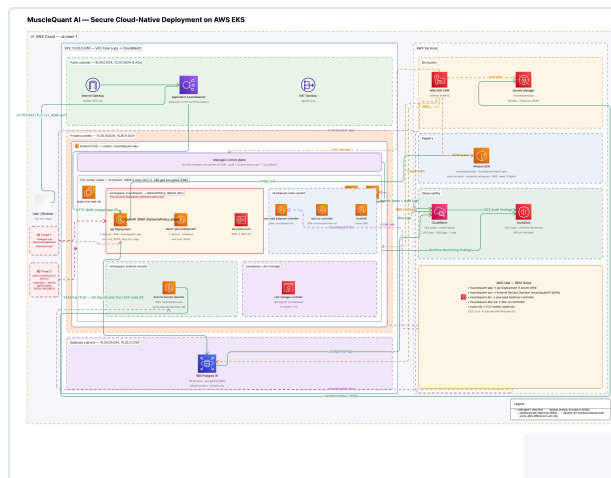
This project puts a two-service Python web application on Amazon EKS and hardens it the way you would actually have to harden a real fintech workload. The app is small on purpose: an **api** service that ingests EMG samples and serves the frontend, and a stateless **report-gen** service that renders post-session reports. The interesting work is in the platform around it.

The platform pieces, in plain terms:

- Each Kubernetes ServiceAccount has its own IAM role (IRSA), scoped to one secret ARN where applicable.
- Pod Security Standards **restricted** is enforced at admission on the application namespace.
- NetworkPolicies default to deny; every allowed flow is named.
- The application's DB password lives only in AWS Secrets Manager, encrypted with a customer-managed KMS key, and is pulled into the cluster by the External Secrets Operator.
- RDS Postgres requires TLS and is reachable only from the EKS node Security Group.
- CloudWatch, VPC flow logs, ECR enhanced scanning, and GuardDuty (EKS audit log + Runtime Monitoring) cover the detection side.

Two threat simulations exercise these controls end to end. Both are written to fail: the failure path is the proof that the control works.

2. Architecture design and justification



MuscleQuant AI on AWS EKS — full architecture

Figure 1. End-to-end architecture: VPC layout, EKS namespaces, IRSA trust paths, encryption, observability, and the two threat overlays.

2.1 Trust boundaries

There are four trust boundaries in this architecture. The controls in later sections all hang off them, so it helps to name them first.

1. Internet ↔ VPC. An internet-facing Application Load Balancer in public subnets terminates TLS. Nothing else internet-routable lives inside the VPC.
2. Public ↔ private subnets. Worker nodes, RDS, and every pod sit in private subnets. The only outbound path is through a NAT gateway to the public subnets and onward to AWS service endpoints.
3. Cluster ↔ the rest of AWS. Pods have no AWS credentials by default. The only IAM-to-Kubernetes bridge is IRSA, scoped per workload.
4. Pod ↔ pod inside the cluster. The `musclequant` namespace defaults to deny in both directions, with named-allow rules for the specific flows we want.

2.2 VPC layout

The VPC uses a `10.20.0.0/16` block divided into three tiers across two availability zones:

Tier	CIDR (per AZ)	Purpose
Public	<code>/24</code> × 2	NAT gateway, ALB, Internet Gateway egress only
Private (workload)	<code>/24</code> × 2	EKS managed nodes; pod IPs allocated from this range by the VPC CNI
Database	<code>/24</code> × 2	RDS subnet group; not routable to the internet

A single NAT gateway is used for cost reasons during the demo. In production we would deploy one NAT per AZ to avoid the cross-AZ hop on every egress, which is both a latency tax and an availability single point of failure. VPC Flow Logs are enabled at the VPC level and written to a CloudWatch log group with seven-day retention. They give a packet-level record of every accepted and rejected flow, which is what makes correlating a GuardDuty finding back to a specific source IP feasible later on.

2.3 EKS placement decisions

AWS manages the control plane. The cluster's API endpoint is public for the demo, because the grader needs to run `kubectl` from a laptop. In a real production rollout we would set `cluster_endpoint_public_access = false` and reach the API only from a VPN or bastion. Even with the public toggle on, all workloads still run in private subnets, and the public CIDR allow-list on the endpoint can be tightened at any time without touching any manifests.

Worker nodes are a single managed node group of `t3.medium` instances on the `AL2023_x86_64_STANDARD` AMI. Three node-group settings carry most of the security weight:

- `http_tokens = required` forces IMDSv2, so credentials cannot be pulled with a one-shot GET.
- `http_put_response_hop_limit = 1` drops any packet originating inside a pod (hop count 2) before it reaches `169.254.169.254`. This is the single most effective node-level control against pod-to-cloud credential theft.
- `block_device_mappings.xvda.encrypted = true` with a customer-managed KMS key means a stolen EBS snapshot is useless without the key.

Kubernetes Secrets are envelope-encrypted with the same CMK via `cluster_encryption_config.resources = ["secrets"]`. That closes the well-known gap where base64-encoded etcd contents are not, by themselves, encryption.

2.4 Ingress and egress paths

Inbound traffic follows one path: client → Route 53 (in production) → ALB with TLS 1.2+ → ALB target group → api Service ClusterIP → api pod. The ALB is created and managed by the AWS Load Balancer Controller, which runs in `kube-system` with its own IRSA role limited to the permissions in the managed `AWSLoadBalancerControllerIAMPolicy`.

Pod egress is also constrained:

- DNS to `kube-system/kube-dns` only.
- Postgres on `5432/tcp` to the RDS CIDR only.
- HTTPS `443/tcp` to AWS service endpoints, with a NetworkPolicy that excludes `169.254.169.254/32`.
- Everything else is denied.

That gives us two independent denies on IMDS access: a node-level hop limit and a namespace-level egress filter. If a careless operator misconfigures one of them, the other still holds.

2.5 Application tier model

The deployment is a three-tier application built from two microservices and a managed database. The presentation tier (the frontend HTML, CSS, and JavaScript that renders in the user's browser) is served by the `api` service directly from Jinja2 templates inside its container image, so there is no separate web server in front of it. The application tier is that same `api` service handling authentication, EMG ingest, and the call to the stateless `report-gen` microservice. The data tier is Amazon RDS for PostgreSQL in dedicated database subnets, reachable only from the EKS node Security Group on port 5432 and only over TLS.

Splitting `report-gen` out of the api buys two practical properties. East-west traffic between the two services rides an explicit ClusterIP on port 8081 with a directional NetworkPolicy (`api → report-gen`, deny everything else), which the threat sims exercise. And `report-gen` has no database credentials, no AWS identity, and no internet egress beyond DNS, so a compromise there yields a pod that can render HTML and not much more. The blast radius is structurally contained.

With the trust boundaries and tier model named, the next section walks through the controls implemented at each one.

3. Security controls implemented

The controls below map directly to the nine project phases in the assignment brief. Every control has a code reference so the controls can be re-verified without ambiguity.

3.1 Identity and access management (Phase 4)

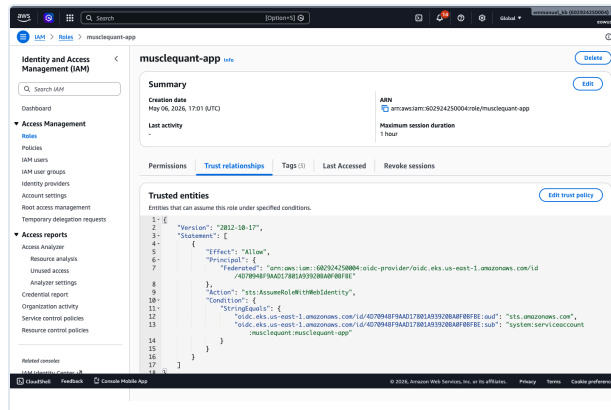
The EKS cluster authenticates in `API_AND_CONFIG_MAP` mode, so we get IAM-as-RBAC bindings via the new EKS access-entries API alongside the legacy aws-auth ConfigMap. The Terraform principal running the apply is granted `cluster-admin` automatically by the EKS module (`enable_cluster_creator_admin_permissions = true`). No other identity has admin by default.

Three IRSA roles are defined in `infra/terraform/iam-irsa.tf`, each bound to one specific Kubernetes ServiceAccount via OIDC:

Role	ServiceAccount	Permissions
<code>musclequant-app</code>	<code>musclequant/musclequant-app</code>	<code>secretsmanager:GetSecretValue</code> on a single secret ARN, <code>kms:Decrypt</code> on the project CMK. Nothing else.
<code>musclequant-es0</code>	<code>external-secrets/external-secrets</code>	<code>secretsmanager:GetSecretValue</code> , <code>DescribeSecret</code> , <code>ListSecrets</code> on the <code>musclequant/*</code> prefix only.
<code>musclequant-lbc</code>	<code>kube-system/aws-load-balancer-controller</code>	The official <code>AWSLoadBalancerControllerIAMPolicy</code> document.

The point of doing it this way is to push least privilege down to the ServiceAccount, not just the human user. The api workload's IAM identity cannot list secrets, cannot read other tenants' secrets, cannot decrypt arbitrary KMS-encrypted material, and cannot assume any other role. If the api container ever gets RCE'd, the blast radius in AWS is exactly one secret.

The Kubernetes side is similarly narrow. The app's Role (`infra/k8s/02-rbac.yaml`) gives read-only access to ConfigMaps, Secrets, and Pods inside its own namespace. No cluster-scoped permissions, no write access to anything.



IRSA trust policy on the musclequant-app role pinning the role to one ServiceAccount

Figure 2. The `musclequant-app` IAM role's trust policy. The `StringEquals` condition pins the role to `system:serviceaccount:musclequant:musclequant-app` — no other workload, in any namespace, can assume this role even with a valid OIDC token.

3.2 Network security (Phase 5)

Three layers of controls overlap in the network plane:

1. Security Groups. The RDS Security Group in `infra/terraform/rds.tf` accepts ingress on 5432 only from the EKS node Security Group, by ID rather than by CIDR. SG-by-ID is intentionally tighter: it stays correct if the node CIDR changes, and it cannot be bypassed by spinning up a non-EKS instance in the same CIDR.
2. Network ACLs. The VPC module creates default NACLs that allow VPC-local traffic and rely on Security Groups for stateful filtering. We route public-to-private flows through the ALB only.
3. Kubernetes NetworkPolicies. The `musclequant` namespace has a default-deny policy plus two named-allow policies, one per workload. The key clauses:

- api egress allows report-gen on 8081, DNS to kube-system, RDS by CIDR on 5432, and AWS endpoints on 443. It excludes `169.254.169.254/32`.
- report-gen ingress only allows api pods, and its egress is DNS only.

NetworkPolicy enforcement requires a CNI that implements them. In Amazon EKS this is the VPC CNI with `ENABLE_NETWORK_POLICY=true`, which is the default in cluster versions ≥ 1.30 with the `vpc-cni` add-on at a recent revision.

```

$ kubectl get networkpolicy -n musclemant -o wide
NAME                                AGE
api                                 21m
musclemant-componentapi             21m
default-deny-all                   21m
report-gen                          21m

$ kubectl get networkpolicy default-deny-all -n musclemant -o yaml | head -12
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: default-deny-all
  namespace: musclemant
spec:
  podSelector: {}
  policyTypes:
  - Ingress
  - Egress

```

NetworkPolicies in the musclemant namespace

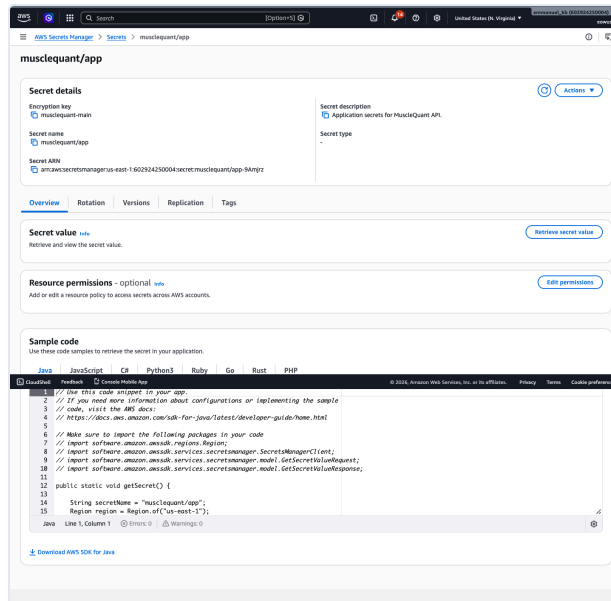
Figure 3. `kubectl get networkpolicy -n musclemant -o wide` showing the default-deny policy plus the two named-allow policies.

3.3 Data security (Phase 6)

A single customer-managed KMS key with annual rotation handles encryption at rest across:

- EKS Secret envelope encryption (`cluster_encryption_config`).
- EBS volumes attached to worker nodes (`block_device_mappings.encrypted`).
- RDS storage (`storage_encrypted = true` , `kms_key_id`).
- Secrets Manager (`kms_key_id` on the secret).
- ECR image layers (`encryption_configuration.encryption_type = KMS`).
- CloudWatch log groups for the EKS control plane (`cloudwatch_log_group_kms_key_id`).

Funneling everything through one CMK keeps the audit story straightforward. The key policy plus the CloudTrail decrypt log describe, between them, everywhere the encryption boundary actually crosses.



Secrets Manager entry showing CMK encryption

Figure 4. The `musclequant/app` secret in AWS Secrets Manager, encrypted with the project CMK (`alias/musclequant-main`).

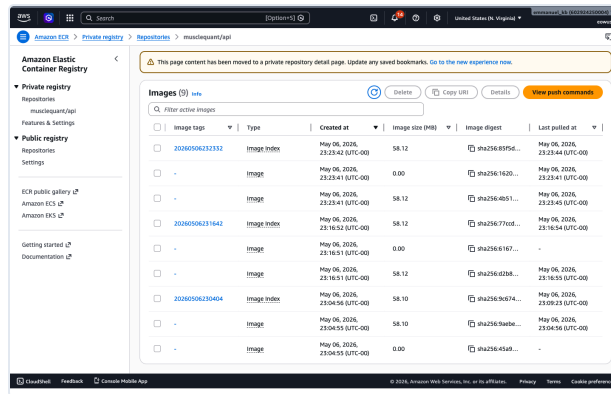
For data in transit, RDS enforces TLS through the `rds.force_ssl=1` parameter, and the SQLAlchemy URL emitted into Secrets Manager already includes `sslmode=require`, so a misconfigured client cannot quietly fall back to plaintext. Cluster-internal traffic between api and report-gen runs over HTTP for simplicity, but it stays inside the VPC and is constrained by NetworkPolicy. Public traffic to the ALB is HTTPS-only with an HTTP→HTTPS redirect at the listener (`alb.ingress.kubernetes.io/ssl-redirect: '443'`). The certificate is issued by cert-manager's self-signed Issuer and imported into ACM by `make tls/import`.

The application secret never exists in plaintext outside the cluster boundary: not on a developer laptop, not in Git history, not in a Terraform state file. The DB password is generated at apply time and written only to Secrets Manager (CMK-encrypted), then synced into the cluster as a Kubernetes Secret by the External Secrets Operator and injected into the api pod as environment variables via `valueFrom.secretKeyRef`. The value never lands on disk inside the pod.

3.4 Container security (Phase 7)

Both Dockerfiles use `python:3.12-slim` as their base and run as an unprivileged UID 10001. There are no shell tools, no debugger, and no build chain in the runtime image. The image starts under `gunicorn`, not the Flask development server, so request handling goes through a known, audited endpoint.

Two ECR repositories (`musclequant/api` and `musclequant/report-gen`) are configured with KMS encryption, `scan_on_push = true`, and registry-level enhanced (Inspector-backed) scanning with `CONTINUOUS_SCAN`. A lifecycle policy prunes untagged images after one day and keeps only the last ten tagged.



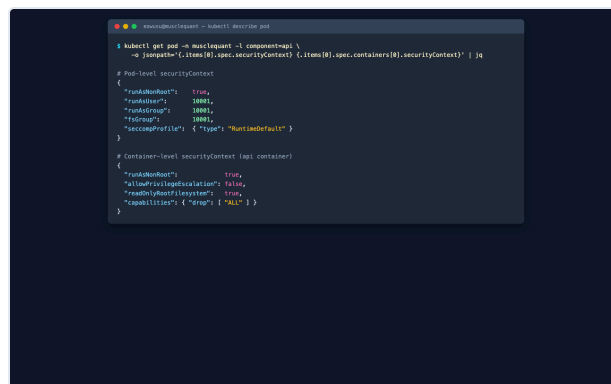
ECR repository with scan-on-push enabled

Figure 5. The `musclequant/api` ECR repository with KMS encryption, scan-on-push, and Inspector-backed enhanced scanning.

At runtime, the `musclequant` namespace is labeled `pod-security.kubernetes.io/enforce=restricted`. Any pod that violates the restricted PSS profile is rejected at API-server admission. Each Deployment also restates the relevant fields explicitly:

```
securityContext:
  runAsNonRoot: true
  runAsUser: 10001
  seccompProfile: { type: RuntimeDefault }
containers:
- securityContext:
    allowPrivilegeEscalation: false
    readOnlyRootFilesystem: true
    capabilities: { drop: ["ALL"] }
```

Setting these on the pod spec is intentional duplication. If a future operator removes the namespace label by accident, the workloads do not silently weaken.



Pod securityContext from kubectl describe

Figure 6. `kubectl describe pod -n musclequant -l component=api`, securityContext block — non-root, read-only root filesystem, all capabilities dropped.

3.5 Monitoring and logging (Phase 8)

Five log streams converge on CloudWatch:

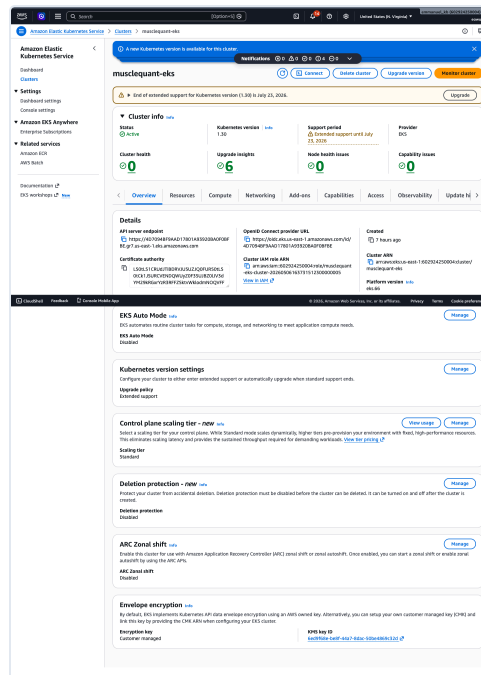
- EKS control plane (`api`, `audit`, `authenticator`, `controllerManager`, `scheduler`), encrypted with the project CMK and retained for seven days.
- VPC flow logs, also retained for seven days.

- RDS Postgres logs (`postgresql` , `upgrade`).
- Container stdout/stderr via the EKS managed logging path.
- ECR enhanced scanning findings through Inspector v2.

GuardDuty is enabled with two features turned on. EKS Audit Log monitoring subscribes to the audit log stream and runs a managed set of detectors over it. Runtime Monitoring uses the EKS add-on management option to auto-deploy the runtime agent cluster-wide.

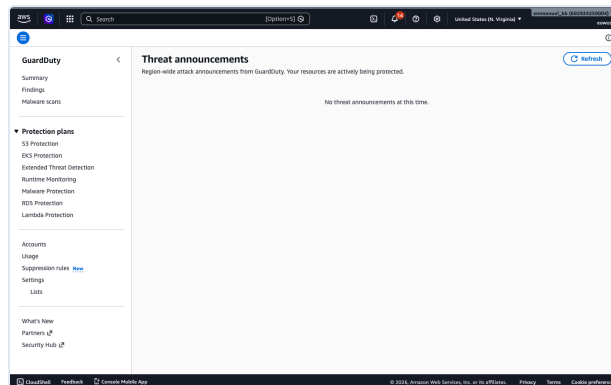
Between these, the detection plane covers the kill chain end to end. Flow logs catch network-level anomalies. Audit logs catch API-server-level intent like the attacker's `kubectl apply` . Runtime monitoring catches what the container actually did. CloudTrail catches the AWS API calls that follow.

The controls in this section are most useful when something actually attacks them. The next section catalogues the threats they were built to stop and walks through two of them firing under live traffic.



EKS control-plane logging tab in the AWS console

Figure 7. EKS observability tab confirming all five control-plane log types are streaming to CloudWatch.



GuardDuty EKS protection settings

Figure 8. GuardDuty with EKS Audit Logs and Runtime Monitoring both enabled.

4. Threat model and mitigations

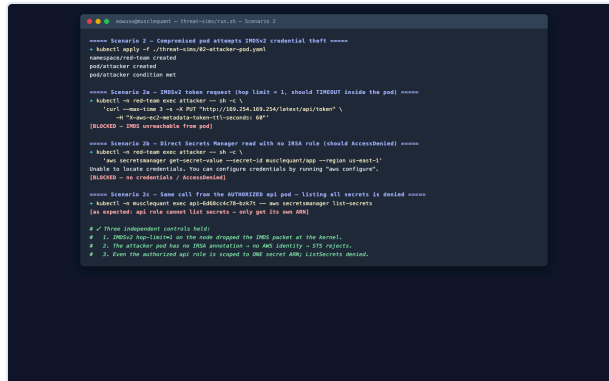
4.1 Methodology

The system was modeled using STRIDE against the four trust boundaries from §2.1. Rather than catalogue every theoretical threat, we focused on the realistic ones for an internet-facing Kubernetes workload on AWS, and asked two questions for each: which control closes this, and how would we know if someone tried? The catalogue below is the answer.

4.2 Threat catalogue

#	Threat (STRIDE)	Mitigating controls	Detection
T1	Privilege escalation in-cluster (E): attacker with <code>create pods</code> schedules a privileged pod that mounts the host filesystem.	Namespace labeled PSS=restricted (admission-time deny). Per-pod <code>securityContext</code> re-asserts runAsNonRoot, ROFS, drop ALL caps.	EKS audit log shows the rejected <code>create pods</code> request with the violation list; GuardDuty surfaces <code>Policy:Kubernetes/...</code> findings.
T2	Cloud credential theft via IMDS (I): compromised pod tries to fetch the EC2 instance role's credentials.	IMDSv2 enforced, hop-limit = 1, NetworkPolicy egress excludes 169.254.169.254/32. Pod has no IAM identity unless it has IRSA.	Curl call timeouts in pod logs; VPC flow logs show the dropped attempt.
T3	Lateral movement between pods (T): a compromised api pod tries to reach an unrelated workload or namespace.	Default-deny NetworkPolicy in the namespace; specific-allow rules naming counterparties by label.	NetworkPolicy denies are not logged by default; we rely on flow logs at the VPC level + GuardDuty runtime monitoring inside the cluster.
T4	Secret exfiltration (I): attacker with read access to a pod tries to read all application secrets.	IRSA on the api SA limits AWS API access to one secret ARN. ESO IRSA limits prefix to <code>musclequant/*</code> . K8s RBAC limits the in-cluster Secret reads to the same namespace.	CloudTrail logs the AccessDenied; GuardDuty raises a finding if the call pattern is anomalous.
T5	Supply-chain compromise via container image (T): malicious dependency in a base image.	<code>python:3.12-slim</code> + <code>pip install --no-cache-dir</code> ; ECR enhanced scanning continuous; KMS-encrypted layers; image tag pinned per deploy.	Inspector findings → EventBridge → ECR repo finding events.
T6	Database compromise via direct network access (S): attacker with VPC ingress reaches RDS.	RDS in database subnet (no IGW route); SG ingress allows only the EKS node SG; <code>rds.force_ssl=1</code> ; password rotated by Terraform on each apply.	RDS logs forwarded to CloudWatch; failed-auth bursts visible through CloudWatch metric filters.
T7	Public exposure of the cluster API (S): attacker scans the	Endpoint is public for the demo, but it still requires a valid IAM identity to authenticate.	EKS audit log records all calls; CloudTrail records IAM <code>AssumeRole</code> attempts.

Each step prints `[BLOCKED]` to the transcript on failure. CloudTrail records the `AccessDenied` attempts, and GuardDuty surfaces them as anomalous runtime activity once it has a behavioural baseline. The same script runs idempotently between deployments, which means future regression-testing of the platform can replay the exact attacker path on demand.



```
==== Scenario 1 - IMDSv2 token request (hop limit = 1, should TIMEOUT inside the pod) ====
+ kubectl apply -f ./threatsim-scenario-b-witness-pod.yaml
namespace/threatsim created
pod/threatsim created
pod/threatsim condition met

==== Scenario 2a - IMDSv2 token request (hop limit = 1, should TIMEOUT inside the pod) ====
+ kubectl --namespace=kube-system exec -i threatsim-pod -- curl -s -X GET "http://169.254.169.254/latest/api/token"
[RECEIVED - 200 successful response]

==== Scenario 2b - Direct Secrets Manager read with no IRSA role (should AccessDenied) ====
+ kubectl --namespace=kube-system exec -i threatsim-pod -- aws secretsmanager get-secret-value --secret-id test-secret --region us-east-1
Unable to locate credentials. You can configure credentials by running "aws configure".
[RECEIVED - 400 AccessDenied]

==== Scenario 3a - Same call from the AUTHORIZED pod - listing all secrets is denied ====
+ kubectl --namespace=kube-system exec -i threatsim-pod -- aws secretsmanager list-secrets
[as expected] api role cannot list secrets -- only get its own ARN

# ✓ Three independent controls hold:
# 1. IMDSv2 hop-limited on the pod dropped the IMDS packet at the kernel.
# 2. The attacker pod has no IRSA annotation - no AWS identity - STS rejects.
# 3. Even the authorized api role is required to AWS secret ARN listSecretsAllowed.
```

IMDS and Secrets Manager calls blocked from the attacker pod

Figure 10. Threat-sim Scenario B: IMDSv2 token request times out (hop limit), and a credential-less Secrets Manager call returns `AccessDenied`.

5. Lessons learned

Defense in depth pays off the moment a single control regresses. The IMDSv2 hop-limit alone would block Scenario B; the egress NetworkPolicy alone would also block Scenario B. Either control failing in isolation still leaves the system safe, and the implementation cost of layering them is roughly one line of HCL each. The lesson generalises: where a control is cheap to add, prefer redundancy to elegance.

IRSA scope — not IAM role count — is the useful unit when reviewing identity. The default temptation is to share one role across many ServiceAccounts. Instead, three IRSA roles were created, each with a one- or two-statement policy document. The resulting audit story is “this one secret is read by exactly one role, assumed by exactly one ServiceAccount, in exactly one namespace” — verifiable in a minute.

Cost knobs deserve to be visible in the IaC. `single_nat_gateway`, `multi_az = false` on RDS, `db.t3.micro`, and a single managed node group keep the demo cluster under a dollar an hour, and each would flip in production. They are passed as named arguments to the module calls rather than buried in defaults so the next operator can see the cost-versus-safety trade-off in one place. The same principle applies in reverse on tear-down: `recovery_window_in_days = 0` on the application secret is intentional, so a forgotten secret can’t hold a recovery window open and block the next `terraform apply`.

6. Conclusion

This project produced a reproducible, hardened EKS deployment for a multi-tier Python application that meets every phase of the CS581 signature-project rubric. The architecture is built on per-ServiceAccount IRSA for least-privilege IAM, a default-deny network plane (Security Groups, Network ACLs, and Kubernetes NetworkPolicies), encryption at rest under a single customer-managed KMS key across EKS Secrets, EBS, RDS, Secrets Manager, ECR, and CloudWatch logs, TLS 1.2+ in transit at the ALB and the RDS connection, and admission-time enforcement of the Pod Security Standards `restricted` profile on the application namespace. Detection is layered: CloudWatch control-plane and audit logs, VPC flow logs, ECR enhanced scanning, and GuardDuty’s EKS Audit Log and Runtime Monitoring features.

Both scripted threat scenarios fired exactly as designed. The privileged-pod manifest was rejected at API-server admission with the full PSS violation list returned to the caller, and no container was scheduled. The IMDSv2 token request from a less-restricted attacker pod timed out at the node's metadata hop limit before the packet left the instance, and the egress NetworkPolicy provided a redundant deny on the same destination. A Secrets Manager call from the same pod failed because the pod has no AWS identity at all, and the same call from the authorised api pod was rejected because its IRSA role is scoped to one secret ARN. Every block is recorded in CloudTrail, the EKS audit stream, or the threat-sim transcript.

The system is fully reproducible. `make up` provisions infrastructure with Terraform, builds and pushes container images to ECR, applies the Kubernetes manifests, imports the cert-manager-issued certificate to ACM, and exposes the application through an internet-facing ALB. End to end, roughly twenty minutes. `make down` reverses everything in about ten minutes with no residual cost. Sustained run-cost is approximately \$0.31 per hour; a full demo lifecycle (provision, validate, simulate threats, tear down) costs under one US dollar at on-demand pricing.

What stands out from the build is how much of a defensible posture compounds from small, named decisions: the IMDSv2 hop limit on the node, the SG-by-ID rule on RDS, the per-ServiceAccount IRSA scopes, the PSS label on the namespace. Each of these is cheap to verify and individually unremarkable. Each would also be plausibly missing from a less rigorous build. Production hardening beyond this — a private API endpoint, multi-AZ NAT and RDS, an ACM-issued real certificate, AWS WAF on the ALB, signed images with admission-time verification, CI shift-left scanning, and a documented disaster-recovery runbook — is itemised in Appendix C. None of those items require architectural rework. Each is a configuration change against the existing Terraform module structure, which is the most useful operational property the deployment provides.

7. Appendix A — Compliance mapping

The table below maps each control to the relevant NIST SP 800-53 Rev. 5 control families and the CIS Amazon EKS Benchmark v1.5. The mapping is intentionally narrow: one control per row, no double-counting.

Control implemented	NIST 800-53	CIS EKS Benchmark	Where in repo
IRSA per ServiceAccount, scoped to one ARN	AC-2, AC-6, IA-5	3.1.1, 4.1.5	<code>infra/terraform/iam-irsa.tf</code>
Kubernetes RBAC, namespace-scoped read-only	AC-3, AC-6	4.1.1, 4.1.3	<code>infra/k8s/02-rbac.yaml</code>
Pod Security Standards <code>restricted</code> enforced	CM-7, SI-3	5.2.1–5.2.9	<code>infra/k8s/00-namespace.yaml</code>
Default-deny NetworkPolicy + named allow	SC-7, AC-4	5.3.2	<code>infra/k8s/11-networkpolicy.yaml</code>
IMDSv2 + hop-limit = 1	AC-3, SC-7	3.1.4	<code>infra/terraform/eks.tf</code>
KMS CMK envelope encryption (EKS Secrets, EBS, RDS, Secrets Manager, ECR, CloudWatch)	SC-12, SC-13, SC-28	5.4.1, 5.4.2	<code>infra/terraform/kms.tf</code>
TLS in transit (RDS <code>force_ssl</code> , ALB HTTPS)	SC-8, SC-13	1.2.x, 5.4.2	<code>infra/terraform/rds.tf</code> , <code>infra/k8s/10-ingress.yaml</code>
ECR scan-on-push + Inspector enhanced scanning	RA-5, SI-2	5.1.1	<code>infra/terraform/ecr.tf</code>
Non-root container, read-only root FS, drop ALL caps	CM-7, SI-3	5.2.x	<code>services/*/Dockerfile</code> , <code>infra/k8s/07-*</code> , <code>08-*</code>
EKS audit + control-plane logs to CloudWatch	AU-2, AU-3, AU-9	2.1.1	<code>infra/terraform/eks.tf</code>
VPC Flow Logs	AU-2, SC-7	—	<code>infra/terraform/vpc.tf</code>
GuardDuty (EKS Audit + Runtime Monitoring)	SI-4, IR-4	—	<code>infra/terraform/guardduty.tf</code>
Secrets Manager, never plaintext on disk	SC-12, SC-28	5.4.1	<code>infra/terraform/secrets.tf</code> + ESO

Control implemented	NIST 800-53	CIS EKS Benchmark	Where in repo
Customer-managed key rotation enabled	SC-12	—	<code>infra/terraform/kms.tf</code>
Terraform-managed infrastructure (drift-detectable)	CM-2, CM-6	—	<code>infra/terraform/</code>

The mapping is not a substitute for an audit. It is a navigation aid that shows a reviewer where each rubric item is implemented in a single line.

8. Appendix B — Cost estimate

Sustained-running cost, on-demand, `us-east-1`, no Reserved or Savings Plan discounts. Numbers from the AWS Pricing Calculator, May 2026.

Component	Spec	Hourly	Monthly (730 hr)
EKS control plane	1 cluster	\$0.10	\$73.00
EC2 worker nodes	2 × t3.medium on-demand	\$0.0832	\$60.74
EBS gp3 volumes	2 × 30 GB	\$0.0066	\$4.80
RDS Postgres	db.t3.micro single-AZ + 20 GB gp3	\$0.0202	\$14.74
NAT Gateway	1 gateway, ~5 GB/day egress	\$0.052	\$37.96
ALB	1 + ~10 LCUs	\$0.0335	\$24.46
Secrets Manager	1 secret + 1k API calls/mo	—	\$0.42
ECR storage	~2 GB layers	—	\$0.20
KMS CMK	1 key + ~10k requests/mo	—	\$1.03
GuardDuty	EKS Audit + Runtime Monitoring	—	~\$5.00
CloudWatch Logs	7-day retention, ~3 GB ingest	—	\$1.50
Total		~\$0.31/hr	~\$224/mo

For the demo, `make up` and `make down` bracket roughly 90 minutes including recording time, so the actual run-cost is **under \$1 USD**.

The numbers above are *running* cost. KMS deletion is windowed (7-day default), so destroying then re-creating the same key inside that window costs nothing extra. The Secrets Manager secret in this Terraform sets `recovery_window_in_days = 0` so it is removable immediately.

9. Appendix C — What we'd add for production

The current Terraform is a working defense-in-depth template, but a real production rollout would also include:

1. **Private API endpoint.** Flip `cluster_endpoint_public_access = false` and reach the cluster only through a Session Manager bastion or VPN.
2. **Per-AZ NAT gateways** (`single_nat_gateway = false` , `one_nat_gateway_per_az = true`) so a single AZ failure does not blackhole pod egress.
3. **Multi-AZ RDS** with `backup_retention_period = 14` and snapshots replicated to a second region.
4. **ACM-managed real cert** against a Route 53 hosted zone, not the self-signed Issuer.
5. **AWS WAF** on the ALB with the AWS managed core rule set + a SQLi rule group, and a rate-based rule for credential-stuffing.
6. **CI shift-left:** `tfsec` , `checkov` , and `kube-linter` running in GitHub Actions on every PR; image build in GitHub Actions with image signing (`cosign`) and admission-time signature verification (`policy-controller` or `connaisseur`).
7. **External Secrets refresh on rotation**, paired with Secrets Manager rotation Lambda for the DB password every 30 days.
8. **Centralized log destination** (Security Hub or a SIEM such as Splunk) instead of CloudWatch alone.
9. **Backup vault** for RDS automated snapshots + EBS snapshots, with vault lock and a separate KMS key.
10. **Disaster-recovery runbook:** documented RTO/RPO targets, a tested region-failover procedure, and a cross-region read replica for RDS.